Quantum Mechanics

Lecture 5: Fourier Duality, Wave Packets, and Photon Polarization

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Chapter 1

Fourier Duality, Wave Packets, and Photon Polarization

1.1 A Periodic Line as a Regulator

Begin with a line segment of length L, but identify its two ends. A particle that leaves at one end reappears at the other. This is equivalent mathematically to motion on a circle, although we need not imagine the particle literally travelling around a physical ring. The construction is a regulator: it makes momentum discrete and normalization ordinary, and the infinite line is recovered by taking L to infinity.

The endpoint identification requires

$$\psi(x + L) = \psi(x). \tag{1.1}$$

It is convenient to write $L = 2\pi r$. The symbol r then resembles a circle radius, but only the circumference $2\pi r$ matters.

Momentum eigenfunctions are plane waves. Restoring units before simplifying the notation,

$$e^{ipx/\hbar} = e^{ikx}, \qquad k := \frac{p}{\hbar}. \tag{1.2}$$

Thus k is the wave number, while the physical momentum is $p = \hbar k$. On one period, a normalized plane wave is

$$\psi_k(x) = \langle x \,|\, k \rangle_{\text{box}} = \frac{e^{ikx}}{\sqrt{2\pi r}}, \qquad 0 \le x < 2\pi r. \tag{1.3}$$

Question from the audience. Why does the normalization contain a square root? [a]

Response. Normalization is imposed on the absolute square. Since $|e^{ikx}|^2 = 1$,

$$\int_0^{2\pi r} |e^{ikx}|^2 \, dx = 2\pi r.$$

Dividing the amplitude by $\sqrt{2\pi r}$ divides its absolute square by $2\pi r$, leaving unit norm.

[a] Lecture timestamp: 00:04:29–00:05:26.

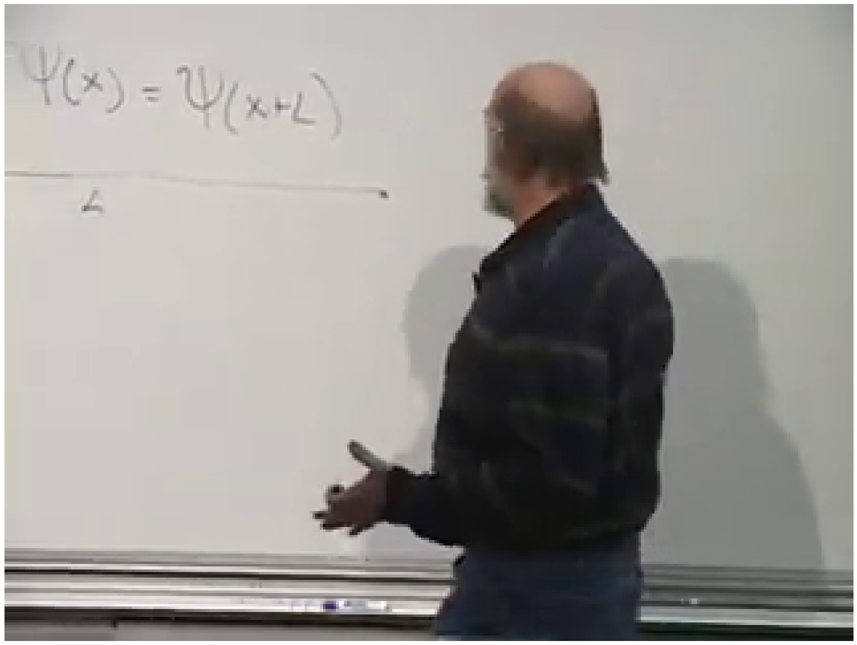

Figure 1.1: The periodicity condition and the finite interval of length L.

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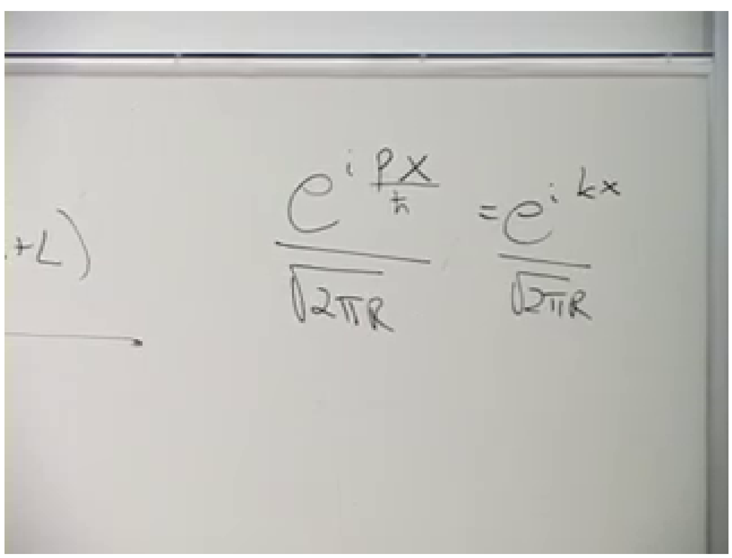

Figure 1.2: The normalized plane wave written in $p/\hbar$ and k notation.

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Periodicity now quantizes the wave number:

$$e^{ik(x+2\pi r)} = e^{ikx}, \tag{1.4}$$

$$e^{i2\pi rk} = 1, \tag{1.5}$$

$$k = \frac{n}{r}, \qquad n \in \mathbb{Z}. \tag{1.6}$$

The allowed values are separated by

$$\Delta k = \frac{1}{r}. \tag{1.7}$$

As r grows, the spectrum becomes dense. A dense quantum spectrum is not by itself a classical limit; it is only the continuum limit of this kinematical basis.

1.2 Orthogonality Before the Continuum

The momentum operator is Hermitian on periodic functions with the appropriate domain, so eigenfunctions with different eigenvalues are orthogonal. Here the theorem can be checked directly. For $k_n = n/r$ and $k_m = m/r$,

$$\langle k_m \mid k_n \rangle_{\text{box}} = \frac{1}{2\pi r} \int_0^{2\pi r} e^{i(k_n - k_m)x}\, dx \tag{1.8}$$

$$= \delta_{mn}. \tag{1.9}$$

When $m = n$, the integrand is one. When $m \neq n$, it completes an integer number of oscillations, and its positive and negative contributions cancel.

Question from the audience. Is orthogonality here a definition or a result? [a]

Response. It is a result. Distinct eigenvalues of a Hermitian operator have orthogonal eigenvectors. The integral above verifies that general theorem for the periodic plane waves; it does not define orthogonality.

[a] Lecture timestamp: 00:11:24–00:12:54.

The states $|k_n\rangle_{\text{box}}$ therefore form an orthonormal basis. At finite r, the correct language is discrete: sums over n and Kronecker deltas δ_{mn}. We now stretch this basis into the continuum without losing its normalization bookkeeping.

1.3 From Kronecker to Dirac Normalization

Because $\Delta k = 1/r$, define continuum-normalized states at the sampled wave numbers by

$$|k_n\rangle := \frac{1}{\sqrt{\Delta k}}\, |k_n\rangle_{\text{box}} = \sqrt{r}\, |k_n\rangle_{\text{box}}\,. \tag{1.10}$$

Their position-space kernels are

$$\langle x \mid k_n \rangle = \sqrt{r}\, \frac{e^{ik_n x}}{\sqrt{2\pi r}} = \frac{e^{ik_n x}}{\sqrt{2\pi}}. \tag{1.11}$$

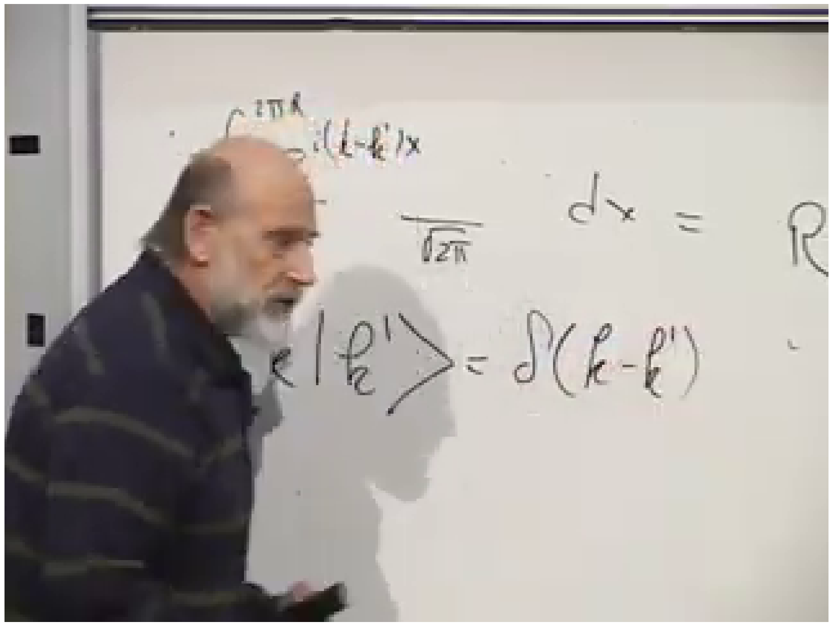

Figure 1.3: The board transition from normalized plane waves to $\delta(k-k')$ normalization. *Lecture frame: 00:16:50.*

The factor of $\sqrt{r}$ has not vanished. It has converted a unit-normalized discrete state into a delta-normalized continuum state. Indeed,

$$\langle k_m \,|\, k_n \rangle = \frac{\delta_{mn}}{\Delta k}, \tag{1.12}$$

which tends distributionally to

$$\langle k' \,|\, k \rangle = \delta(k - k'). \tag{1.13}$$

The peak grows like $1/\Delta k = r$ while its width shrinks like Δk, leaving unit area.

Question from the audience. Where did the factor $\sqrt{r}$ go in the overlap? [a]

Response. There is one factor from the bra and one from the ket. Together they multiply the old overlap by r. Thus unequal sampled wave numbers remain orthogonal, while the equal-wave-number overlap grows from 1 to $r = 1/\Delta k$, exactly the height needed for a Dirac delta in the continuum limit.

[a] Lecture timestamp: 00:17:37–00:18:39.

Completeness transforms consistently:

$$\mathbf{1} = \sum_n |k_n\rangle_{\text{box}}\,{}_{\text{box}}\langle k_n| \tag{1.14}$$

$$= \sum_n \Delta k\ |k_n\rangle \langle k_n| \longrightarrow \int_{-\infty}^{\infty} dk\ |k\rangle \langle k| \,. \tag{1.15}$$

The rescaling, the delta function, and the replacement of a sum by an integral are three parts of one operation.

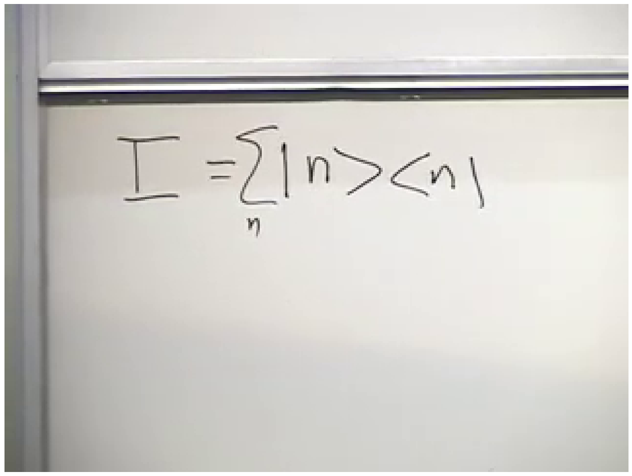

Figure 1.4: A complete orthonormal basis resolves the identity into projectors.
Lecture frame: 00:25:02.

1.4 Dyads and the Resolution of Identity

An inner product such as $\langle a \,|\, b\rangle$ is a number. Reversing the order creates a different kind of object:

$$|b\rangle \langle a| \,. \tag{1.16}$$

This outer product, or dyad, is an operator. Acting on a third vector,

$$(|b\rangle \langle a|)\, |c\rangle = \langle a \,|\, c\rangle \; |b\rangle \,. \tag{1.17}$$

The bra extracts the component of $|c\rangle$ along $|a\rangle$, and the ket places that coefficient along $|b\rangle$.

For any orthonormal basis $\{|n\rangle\}$,

$$|v\rangle = \sum_n v_n \, |n\rangle \,, \qquad v_n = \langle n \,|\, v\rangle \,. \tag{1.18}$$

Substitution gives

$$|v\rangle = \sum_n |n\rangle \langle n| \, |v\rangle \,. \tag{1.19}$$

Since the relation holds for every $|v\rangle$,

$$\mathbf{1} = \sum_n |n\rangle \langle n| \,. \tag{1.20}$$

Question from the audience. Does the repeated symbol n refer to one special state?[a]

Response. No. It is a dummy summation label that runs over every vector in the chosen basis. Another complete orthonormal basis gives another resolution of the same identity. For a continuous label, the sum becomes an integral and Kronecker normalization becomes Dirac normalization.

[a] Lecture timestamp: 00:25:58–00:27:40.

In particular,

$$\mathbf{1} = \int dx \; |x\rangle \langle x| = \int dk \; |k\rangle \langle k| \,. \tag{1.21}$$

These two forms of the identity are the mechanism behind the Fourier transform.

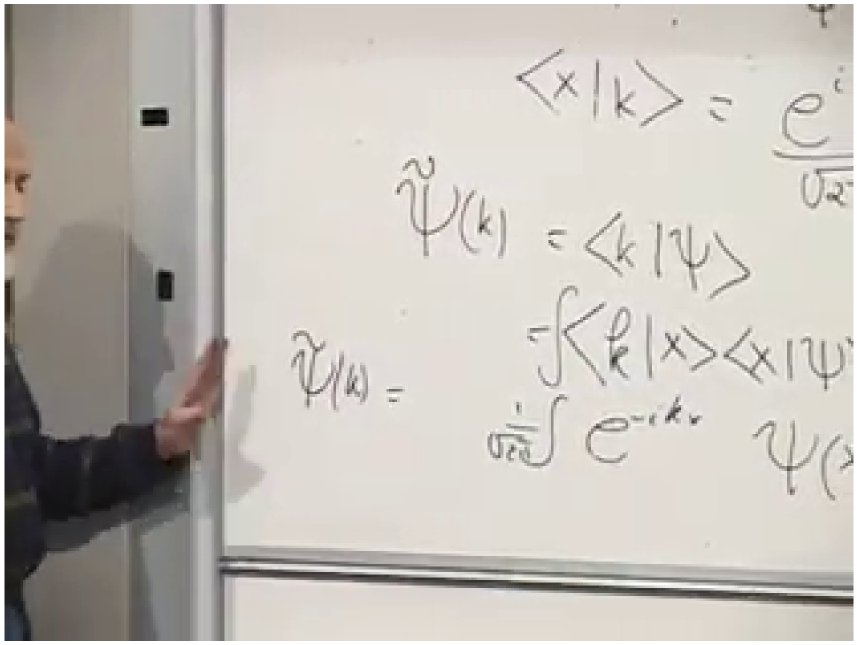

Figure 1.5: Inserting the position basis turns the momentum amplitude into a Fourier integral.
Lecture frame: 00:34:30.

1.5 One State in Two Representations

For an abstract state $|\psi\rangle$, define

$$\psi(x) := \langle x \,|\, \psi \rangle\,, \qquad \widetilde{\psi}(k) := \langle k \,|\, \psi \rangle\,. \tag{1.22}$$

They are not two states. They are the components of one state in two different bases. The change-of-basis kernel is

$$\langle x \,|\, k \rangle = \frac{e^{ikx}}{\sqrt{2\pi}}, \qquad \langle k \,|\, x \rangle = \frac{e^{-ikx}}{\sqrt{2\pi}}. \tag{1.23}$$

Insert the position resolution of identity into $\langle k \,|\, \psi \rangle$:

$$\widetilde{\psi}(k) = \int dx\ \langle k \,|\, x \rangle \langle x \,|\, \psi \rangle \tag{1.24}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, e^{-ikx} \psi(x). \tag{1.25}$$

Insert the momentum resolution into $\langle x \,|\, \psi \rangle$:

$$\psi(x) = \int dk\ \langle x \,|\, k \rangle \langle k \,|\, \psi \rangle \tag{1.26}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk\, e^{ikx} \widetilde{\psi}(k). \tag{1.27}$$

These are the forward and inverse Fourier transforms. Fourier analysis is older than quantum mechanics; quantum mechanics interprets the transform as a change between position and momentum descriptions.

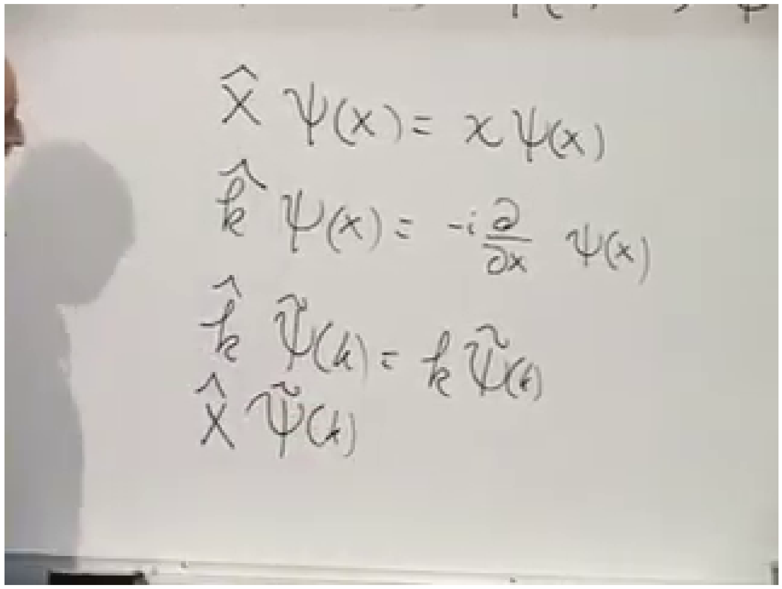

Figure 1.6: Multiplication and differentiation exchange roles in the x- and k-representations.
Lecture frame: 00:42:15.

Question from the audience. Can k be negative, and what does the symmetry between x and k mean physically?[a]

Response. Negative k means momentum in the opposite direction. The deeper symmetry is that position and momentum are conjugate variables. Their reciprocity is natural in Fourier analysis and later appears in classical mechanics through canonical coordinates and Poisson brackets.

[a] Lecture timestamp: 00:38:43–00:39:43.

The operator rules display the same reciprocity. In the position representation,

$$\hat{X}\psi(x) = x\psi(x), \tag{1.28}$$

$$\hat{K}\psi(x) = -i\frac{\partial}{\partial x}\psi(x). \tag{1.29}$$

In the wave-number representation,

$$\hat{K}\widetilde{\psi}(k) = k\widetilde{\psi}(k), \tag{1.30}$$

$$\hat{X}\widetilde{\psi}(k) = i\frac{\partial}{\partial k}\widetilde{\psi}(k). \tag{1.31}$$

The physical momentum operator is $\hat{P} = \hbar\hat{K}$.

Question from the audience. Why is the state represented by $\psi(x)$ or $\widetilde{\psi}(k)$, rather than by one probability amplitude depending on both x and k?[a]

Response. These are two complete representations of the same state, and they are related invertibly by a Fourier transform. Position and momentum are incompatible observables: no normalizable state has both exactly sharp. A definite-k state is spread over all x, while a definite-position state is spread over all k. Quantum mechanics does not assign simultaneous sharp values merely because either representation can be used.

[a] Lecture timestamp: 00:43:13–00:44:36.

Question from the audience. Is the classical conservation of phase-space volume related to quantum time evolution?[a]

Response. Yes, at the structural level emphasized in the lecture. Hamiltonian evolution is a canonical transformation and preserves phase-space volume. It cannot collapse a finite cloud of distinct initial conditions to one point. Quantum evolution is unitary and preserves inner products, so distinctions between state vectors are likewise maintained. The detailed dynamical statement is still ahead of us.

[a] Lecture timestamp: 00:44:38–00:46:39.

Question from the audience. Where does the differential form $\hat{K} = -i\partial_x$ come from?[a]

Response. It is the position-representation form introduced for the momentum generator. Acting on a plane wave,

$$-i\frac{\partial}{\partial x}e^{ikx} = ke^{ikx},$$

so e^{ikx} is an eigenfunction with eigenvalue k. Fourier transformation then gives the reciprocal rule $\hat{X} = i\partial_k$ in momentum space.

[a] Lecture timestamp: 00:46:43–00:48:08.

1.6 Wave Packets and the Classical Bridge

At this stage, the operators have been named position and momentum, but the connection with classical motion has not yet been proved. The bridge is a wave packet. Its magnitude is concentrated near some position x_0, and its Fourier transform is concentrated near some wave number k_0. A typical form is

$$\psi(x) = f(x - x_0)e^{ik_0 x}, \tag{1.32}$$

where f is a smooth localized envelope. The oscillations inside the envelope carry the approximate momentum, while the envelope locates the particle.

Question from the audience. Must the packet be Gaussian? [a]

Response. No. A Gaussian is the cleanest example because its Fourier transform is also Gaussian and it can minimize $\Delta x\,\Delta k \geq 1/2$. The lecture needs only a smooth packet localized in both representations; its exact shape is not essential.

[a] Lecture timestamp: 00:50:23–00:50:34.

Under a sufficiently smooth potential, a narrow packet can retain its integrity for a useful interval. Its center moves through position space, and the center of its momentum distribution moves through momentum space. The theorem promised in the lecture is the Ehrenfest relation: under the conditions in which the packet remains narrow, these expectation values approximately obey the classical equations of motion. The packet may also spread or distort, so the classical description is an approximation, not a new postulate.

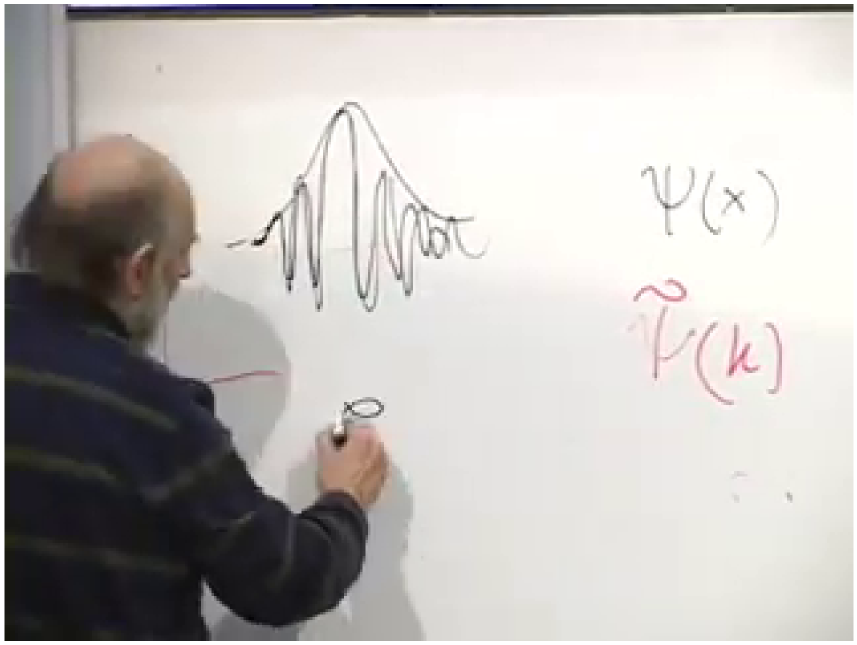

Figure 1.7: A localized envelope modulates a nearly monochromatic plane wave, producing a packet localized in both representations.

Lecture frame: 00:51:25.

Question from the audience. What establishes that this quantum quantity really is classical momentum?[a]

Response. Not the name alone. The connection is established when a localized wave packet evolves according to the Schrodinger equation and its centers in x and $p = \hbar k$ follow the corresponding classical trajectory. That result is announced here and derived only after quantum dynamics has been introduced.

[a]Lecture timestamp: 00:48:13–00:54:02.

1.7 Incompatibility in a Two-State System

For complex vectors, $|A\rangle\langle B|$ is generally not Hermitian, and

$$(|A\rangle\langle B|)^\dagger = |B\rangle\langle A|\,. \tag{1.33}$$

Hermitian conjugation is the operator analogue of complex conjugation.

Question from the audience. What is the outer product of two complex vectors? [a]

Response. It is an operator, usually a non-Hermitian one. Reversing the vectors and converting each ket to its adjoint bra gives its Hermitian conjugate. The special dyad $|A\rangle\langle A|$ is Hermitian and, for normalized $|A\rangle$, is a projector.

[a]Lecture timestamp: 00:54:15–00:55:32.

The larger point is incompatibility. Classically one specifies position *and* momentum. Quantum mechanically one may use the position representation *or* the momentum representation. The state

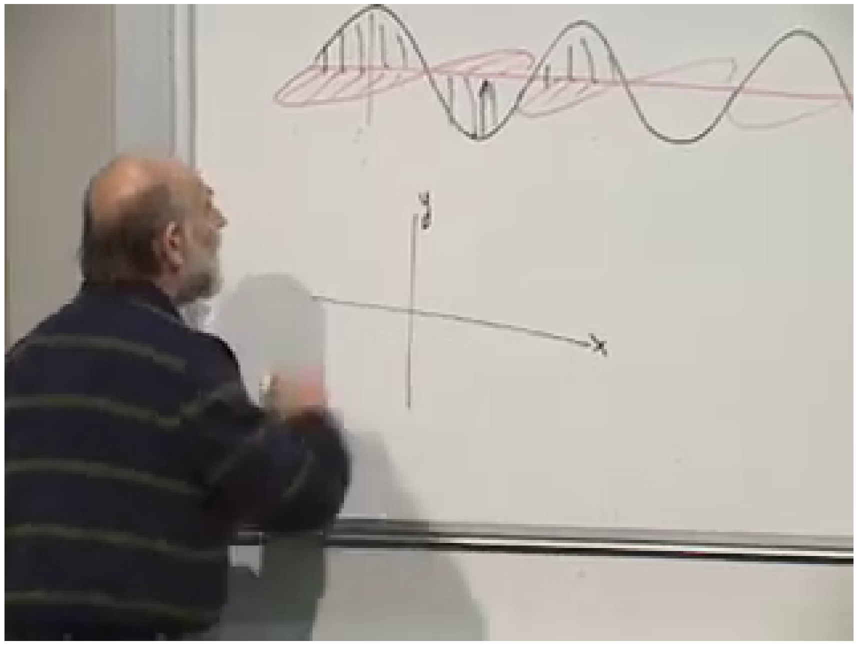

Figure 1.8: Perpendicular electric and magnetic oscillations in a plane-polarized electromagnetic wave.

Lecture frame: 01:00:45.

contains the probability information needed for either experiment, but it does not make incompatible quantities simultaneously definite.

To see the same logic without infinite-dimensional function spaces, turn to photon polarization. Ignore the photon's position and retain only two polarization degrees of freedom.

1.8 From Electromagnetic Waves to a Polarizer

A classical light wave travelling along the z-axis has electric and magnetic fields transverse to its motion and perpendicular to one another. For a plane wave they oscillate in phase. The polarization direction is defined to be the direction in which the electric field oscillates; the opposite direction along the same axis is not a different linear polarization.

A wire-grid polarizer makes this physical. If its conducting wires are vertical, a vertical electric field drives currents in the wires. Those currents reflect or absorb that component. A horizontal electric field cannot drive a horizontal current because no conducting path runs that way, so it is transmitted. Thus the transmission axis is perpendicular to the wires. More generally, an ideal polarizer transmits the electric-field component along one axis and removes the orthogonal component.

The wave description gives continuously variable field amplitudes. The single-photon description gives discrete events. A photon incident on a polarizer is either transmitted or it is absorbed or reflected. Its energy does not split into two fractional photons.

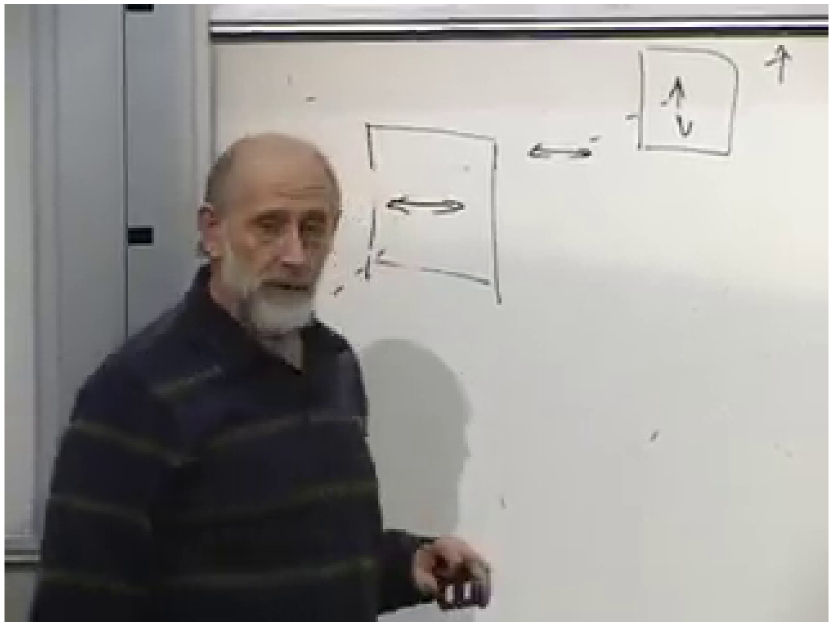

Figure 1.9: Two polarizers illustrate preparation followed by a polarization measurement.
Lecture frame: 01:10:00.

1.9 Preparation and Measurement

A photon that emerges from an ideal vertical polarizer is vertically polarized. If it then meets a second vertical polarizer, it passes with probability one. If it meets a horizontal polarizer, it is blocked with probability one. The first apparatus prepares a state; the second tests it. The same physical device can therefore serve as a state preparer or as a measurement apparatus, depending on its role in the experiment.

Question from the audience. Is a reflected photon itself polarized? [a]

Response. It may be. The lecture discards that output channel and counts only transmission. One may instead build a fuller experiment that detects both channels, but the two-outcome polarization logic for the transmitted beam is unchanged.

[a] Lecture timestamp: 01:06:14–01:06:38.

Choose horizontal and vertical axes called x and y. In this section they label polarization, not the particle's position. Represent the two orthogonal states by

$$|x\rangle_{\text{pol}} \leftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |y\rangle_{\text{pol}} \leftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.34}$$

They satisfy

$$\langle x \,|\, x\rangle = \langle y \,|\, y\rangle = 1, \qquad \langle x \,|\, y\rangle = 0. \tag{1.35}$$

They are orthogonal because a single ideal polarization experiment can distinguish them perfectly.

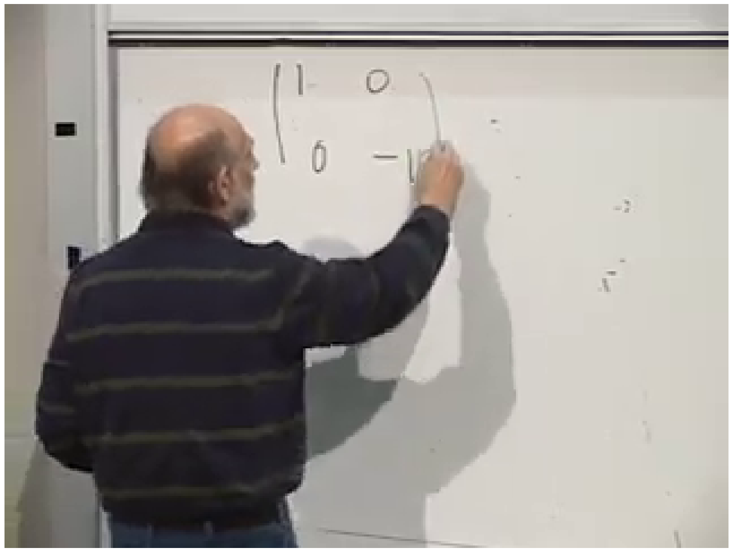

Figure 1.10: The horizontal-vertical polarization observable is diagonal in the $(|x\rangle, |y\rangle)$ basis.
Lecture frame: 01:19:45.

1.10 States, Outcomes, and the Polarization Operator

The state vectors are not themselves observables. To define an observable, associate numerical records with the two possible outcomes. Following the lecture, record $+1$ for x-polarization and -1 for y-polarization. The corresponding operator obeys

$$\hat{P}_{xy}\,|x\rangle = +\,|x\rangle\,, \qquad \hat{P}_{xy}\,|y\rangle = -\,|y\rangle\,. \tag{1.36}$$

In the $(|x\rangle, |y\rangle)$ basis,

$$\hat{P}_{xy} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |x\rangle\langle x| - |y\rangle\langle y|\,. \tag{1.37}$$

Question from the audience. If polarization is observable, why is the state not called an observable?[a]

Response. The state describes the preparation and supplies probability amplitudes. The observable is the Hermitian operator tied to a measurement setup; its eigenvectors are states with definite outcomes, and its eigenvalues are the numbers recorded for those outcomes.

[a] Lecture timestamp: 01:14:56–01:16:03.

Knowing the action of $\hat{P}_{xy}$ on the two basis vectors determines its action on every superposition. This is why specifying eigenvectors and eigenvalues is enough to specify an observable.

1.11 Rotate the Apparatus

Nothing prevents us from rotating a polarizer by 45°. A photon that emerges from it is neither in $|x\rangle$ nor in $|y\rangle$, but an x- or y-oriented test gives either result with equal probability. The normalized

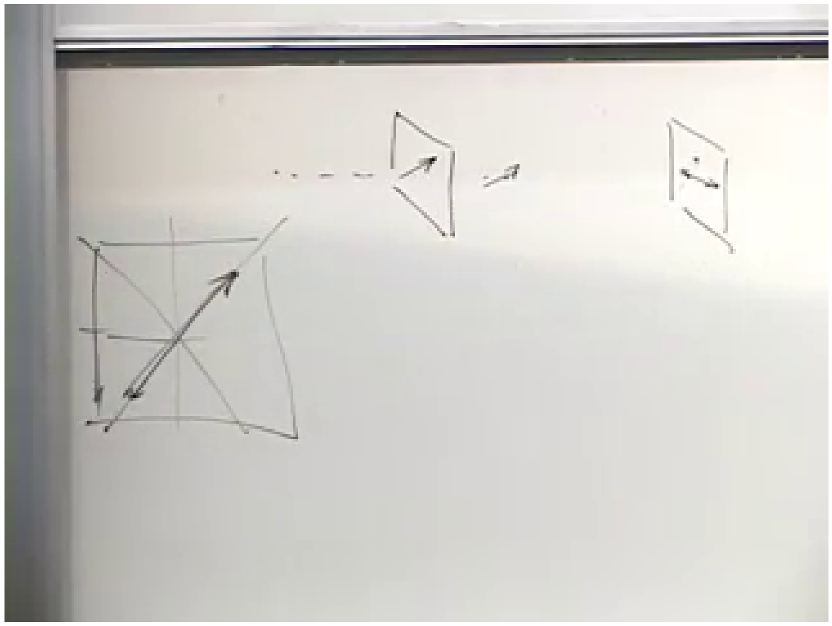

Figure 1.11: A diagonal preparation followed by a horizontal or vertical polarization test.
Lecture frame: 01:23:45.

state is

$$|+\rangle = \frac{|x\rangle + |y\rangle}{\sqrt{2}}. \tag{1.38}$$

The orthogonal linear polarization is

$$|-\rangle = \frac{|x\rangle - |y\rangle}{\sqrt{2}}. \tag{1.39}$$

The relative minus sign makes the two diagonal states orthogonal:

$$\langle - \,|\, + \rangle = \frac{1}{2}\left(\langle x \,|\, x\rangle + \langle x \,|\, y\rangle - \langle y \,|\, x\rangle - \langle y \,|\, y\rangle\right) = 0. \tag{1.40}$$

In column form,

$$|+\rangle \leftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad |-\rangle \leftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}. \tag{1.41}$$

The Born rule confirms the equal probabilities:

$$\text{Prob}(y|+) = |\langle y \,|\, + \rangle|^2 = \left|\frac{1}{\sqrt{2}}\left(\langle y \,|\, x\rangle + \langle y \,|\, y\rangle\right)\right|^2 \tag{1.42}$$

$$= \frac{1}{2}, \tag{1.43}$$

$$\text{Prob}(x|+) = \frac{1}{2}. \tag{1.44}$$

For $|-\rangle$, one amplitude changes sign, but its absolute square is again $1/2$. Amplitudes carry signs and phases; probabilities do not retain them directly.

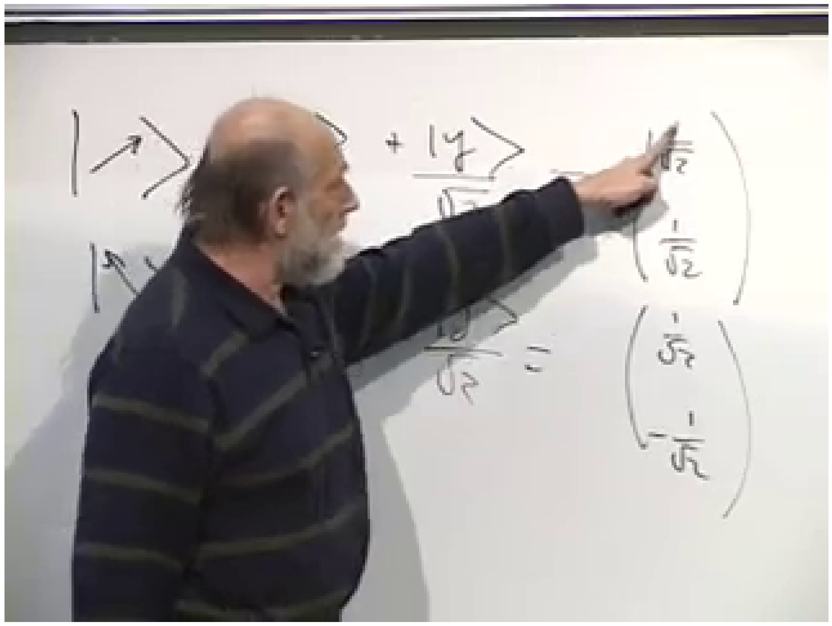

Figure 1.12: The two diagonal polarization states form a second orthonormal basis.
Lecture frame: 01:29:00.

Define the rotated observable by

$$\hat{P}_{45}\,|+\rangle = +\,|+\rangle\,, \qquad \hat{P}_{45}\,|-\rangle = -\,|-\rangle\,. \tag{1.45}$$

In the original x, y basis,

$$\hat{P}_{45} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |+\rangle\,\langle+| - |-\rangle\,\langle-|\,. \tag{1.46}$$

The eigenvectors of $\hat{P}_{xy}$ are $|x\rangle\,, |y\rangle$; those of $\hat{P}_{45}$ are $|+\rangle\,, |-\rangle$. No nonzero vector is an eigenvector of both operators. Equivalently,

$$[\hat{P}_{xy}, \hat{P}_{45}] \neq 0. \tag{1.47}$$

The two polarization observables are incompatible, just as position and momentum are incompatible, but now the entire statement fits inside a two-dimensional vector space.

For an arbitrary linear-polarization angle θ, the announced generalization is

$$|\theta\rangle = \cos\theta\ |x\rangle + \sin\theta\ |y\rangle\,, \tag{1.48}$$

$$|\theta_\perp\rangle = -\sin\theta\ |x\rangle + \cos\theta\ |y\rangle\,. \tag{1.49}$$

The corresponding observable has these two states as eigenvectors. Rotating the axis by 90° merely exchanges the outcomes and reverses the sign of the same observable. Distinct nonorthogonal axes give genuinely incompatible tests. Complex relative coefficients introduce circular and elliptical polarization; that extension is reserved for the next lecture.

1.12 What One Experiment Records

The closing discussion distinguishes an outcome from a probability. A single photon gives one definite record: transmitted or blocked, represented here by $+1$ or -1. The probability of either

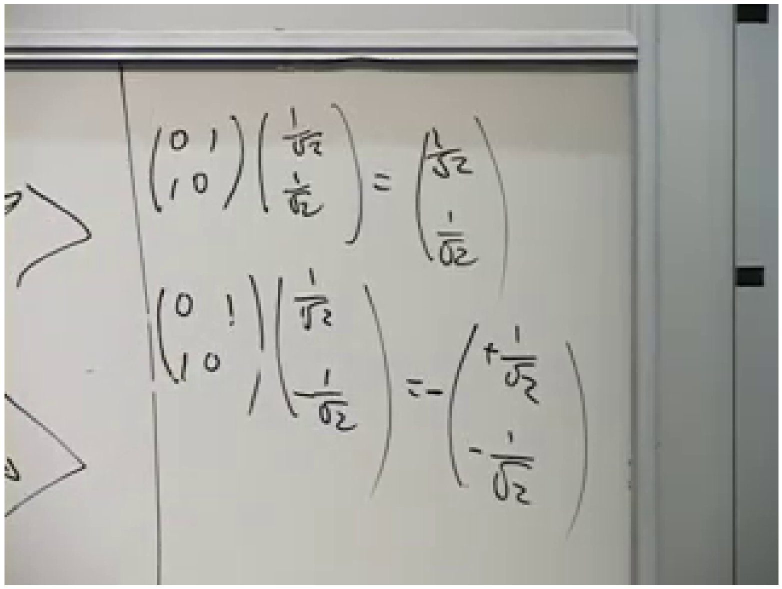

Figure 1.13: The off-diagonal matrix returns $|+\rangle$ and reverses $|-\rangle$, giving the required eigenvalues.
Lecture frame: 01:38:30.

record is inferred from many repetitions with identically prepared photons.

Question from the audience. Is the probability that a photon passes a polarizer itself an observable?[a]

Response. Not in the single-shot technical sense used here. One run produces an outcome, not a probability. Probability is estimated from the relative frequency of outcomes over an ensemble of repeated preparations.

[a]Lecture timestamp: 01:42:27–01:45:03.

The probability for a particular eigenstate $|a\rangle$ is obtained directly from the overlap,

$$\text{Prob}(a|\psi) = |\langle a \,|\, \psi\rangle|^2 . \tag{1.50}$$

One does not first apply the observable to $|\psi\rangle$ and then square an eigenvalue. The observable identifies the possible values and the eigenvectors associated with them; the state's overlaps with those eigenvectors determine the probabilities.

Question from the audience. Should the observable act on the state before the probability is calculated?[a]

Response. No. To find the probability of a particular result, take the inner product of the prepared state with the eigenvector for that result, then square its magnitude. Applying the operator is useful for expectation values and eigenvalue equations, but it is not an extra step in the basic Born-rule calculation.

[a]Lecture timestamp: 01:45:03–01:48:02.

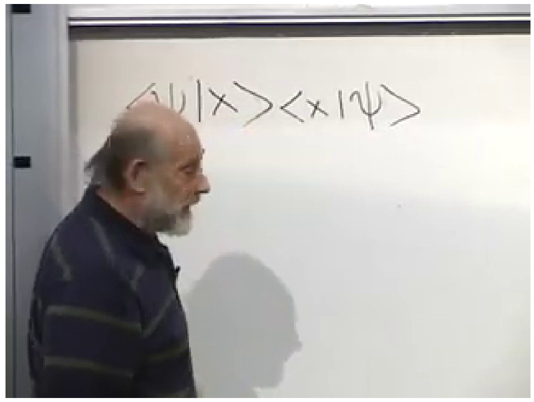

Figure 1.14: A projector expectation gives the x-polarization probability.
Lecture frame: 01:50:30.

Question from the audience. Is an observable synonymous with a Hermitian operator? [a]

Response. Within the finite-dimensional framework of this lecture, yes: observables are represented by Hermitian operators, and every Hermitian operator may be associated with an observable. In infinite-dimensional problems, the precise mathematical condition is self-adjointness with a specified domain.

[a]Lecture timestamp: 01:48:04–01:48:14.

The numerical labels $+1$ and -1 are a convention. We could record 1 and 0, or any two distinct real numbers, provided the convention is used consistently. Choosing 1 and 0 produces the projector

$$\Pi_x = |x\rangle \langle x| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{1.51}$$

For a normalized state $|\psi\rangle$,

$$\begin{aligned} \langle\psi| \Pi_x |\psi\rangle &= \langle\psi| x\rangle \langle x \mid \psi\rangle & (1.52) \\ &= |\langle x \mid \psi\rangle|^2 . & (1.53) \end{aligned}$$

The expectation value equals the probability, but the result of one projector measurement is still 1 or 0.

Question from the audience. If the eigenvalue labels are arbitrary, what does the apparatus actually define?[a]

Response. The apparatus defines the mutually exclusive outcomes and the states that give them with certainty. We then choose real numbers to record those outcomes. With the $+1, -1$ convention the observable is a difference of projectors; with the $1, 0$ convention it is one projector, whose expectation value is the ensemble probability for that outcome.

[a]Lecture timestamp: 01:48:16–01:55:23.

An observable is therefore inseparable from a specified experimental question. The polarizer orientation fixes the eigenvectors, the notebook convention fixes the eigenvalues, and one interaction yields one recorded event. Repetition reveals the probabilities. In this two-state example, the abstract language of vectors, operators, incompatible bases, and Born amplitudes has become a concrete laboratory procedure.